

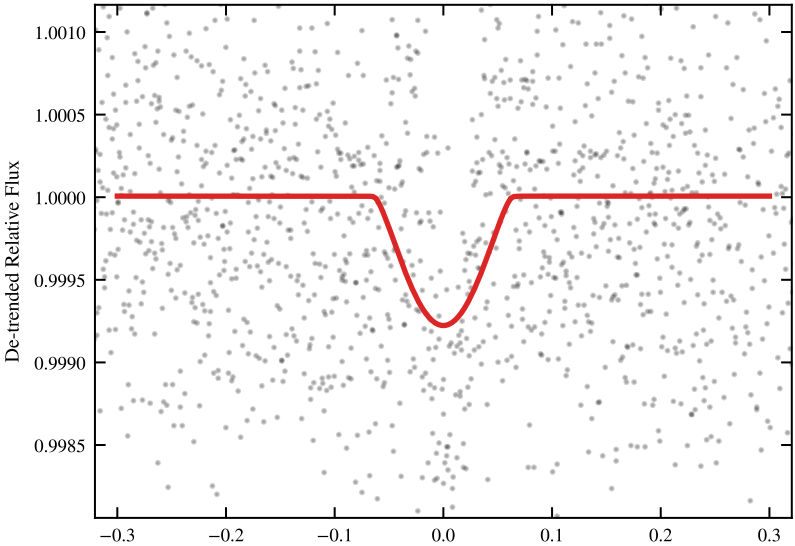
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 319259194 | RA: 71.65078° | Dec: -78.56161° | Sectors: 61, 62, 63, 64, 65, 66, 67, 68

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	70.362 ± 0.00403
Rp/R★	0.27327 ± 0.0989
Rp (R⊕)	62.695 ± 22.9
b	1.2508 ± 0.101
T14 (h)	3.0775 ± 0.202
a (AU)	0.40434 ± 0.0276
Teq (K)	712.96 ± 24.3

Primary Phased Transit



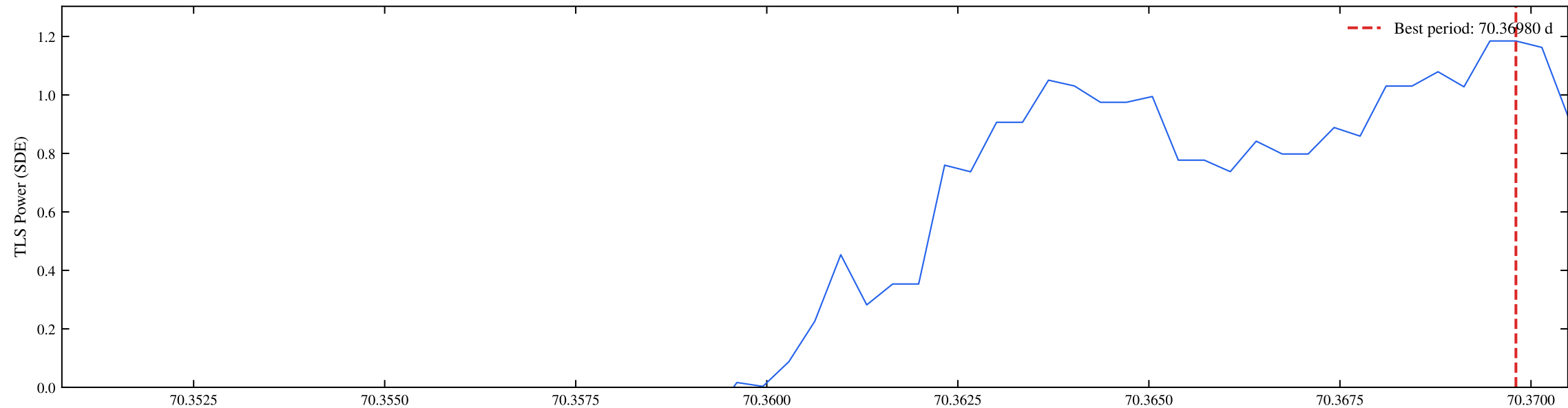
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	70.3698	70.362337 ± 0.00403
P1 Epoch (BTJD)	N/A	2987.5080 ± 0.0050
Max R-hat (R)	N/A	1.0007
Min ESS	N/A	5421
MCMC Divergences	N/A	0

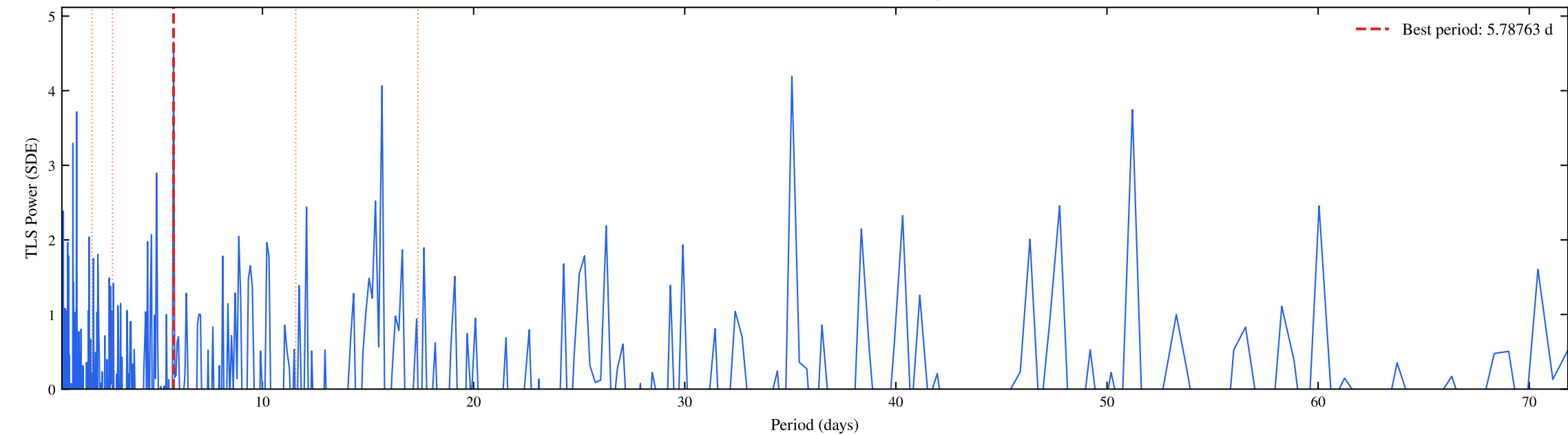
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	2.098 ± 0.106	VizieR TIC8
M★ (M☉)	1.33 ± 0.16	VizieR TIC8
Teff (K)	6495 ± 164	VizieR TIC8
log g (dex)	3.918 ± 0.0777	VizieR TIC8
[Fe/H]	-0.3051 ± 0.037	Gaia DR3
ρ★ (g/cm³)	0.2031 ± 0.0393	Derived

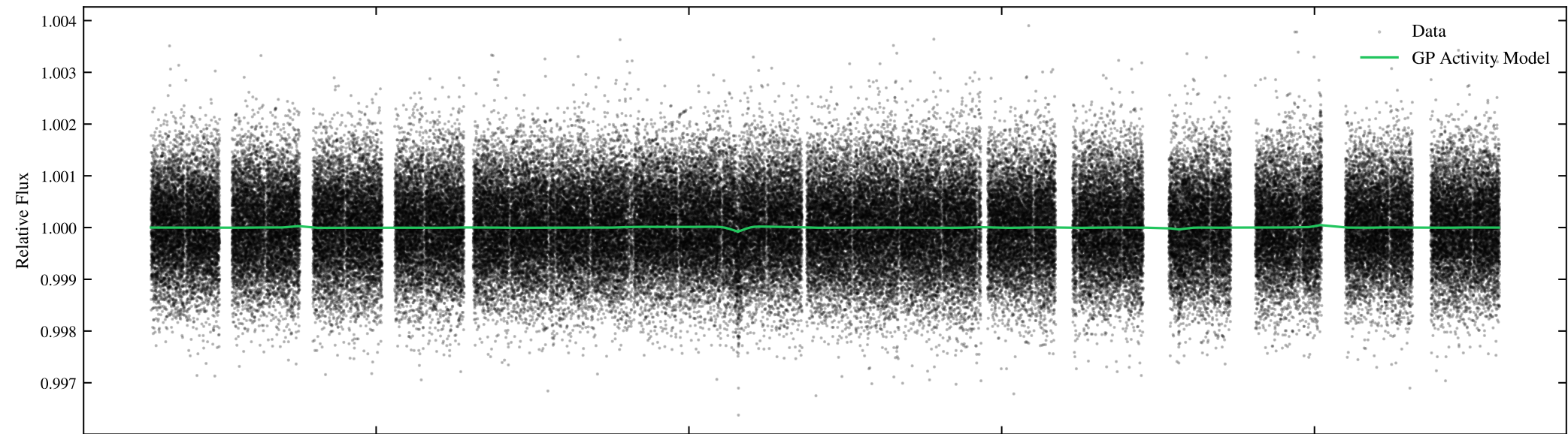
TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | TLS Periodogram (Planet 1 Search) (SDE = 1.18)



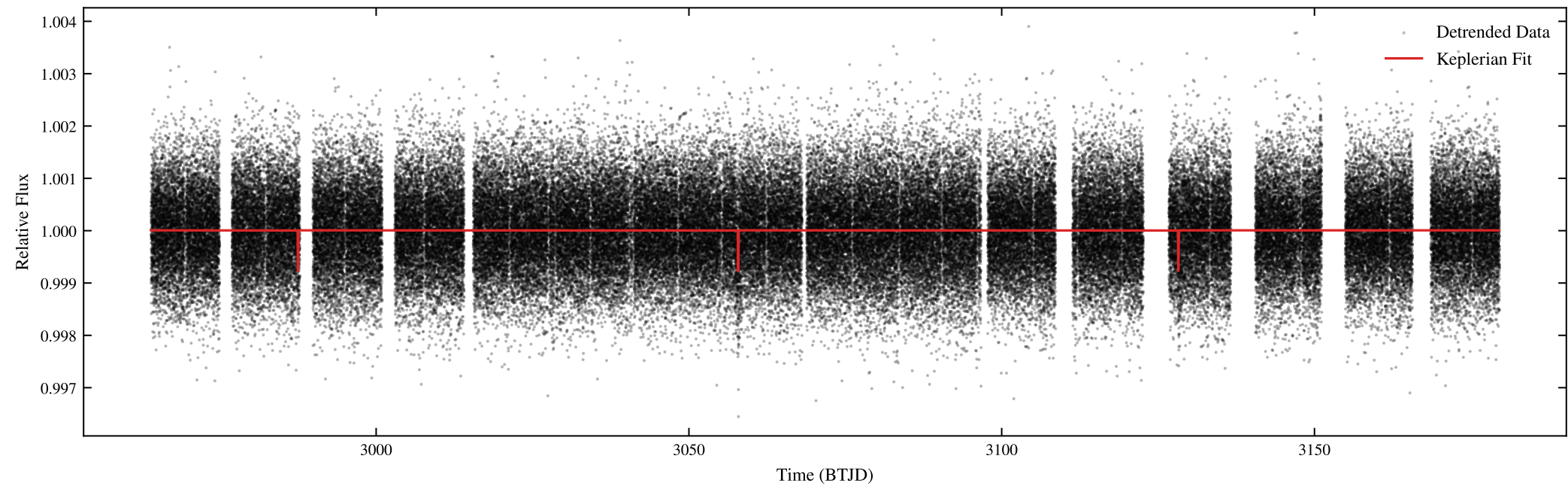
TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | TLS Periodogram (Planet 2 Search) (SDE = 4.65)



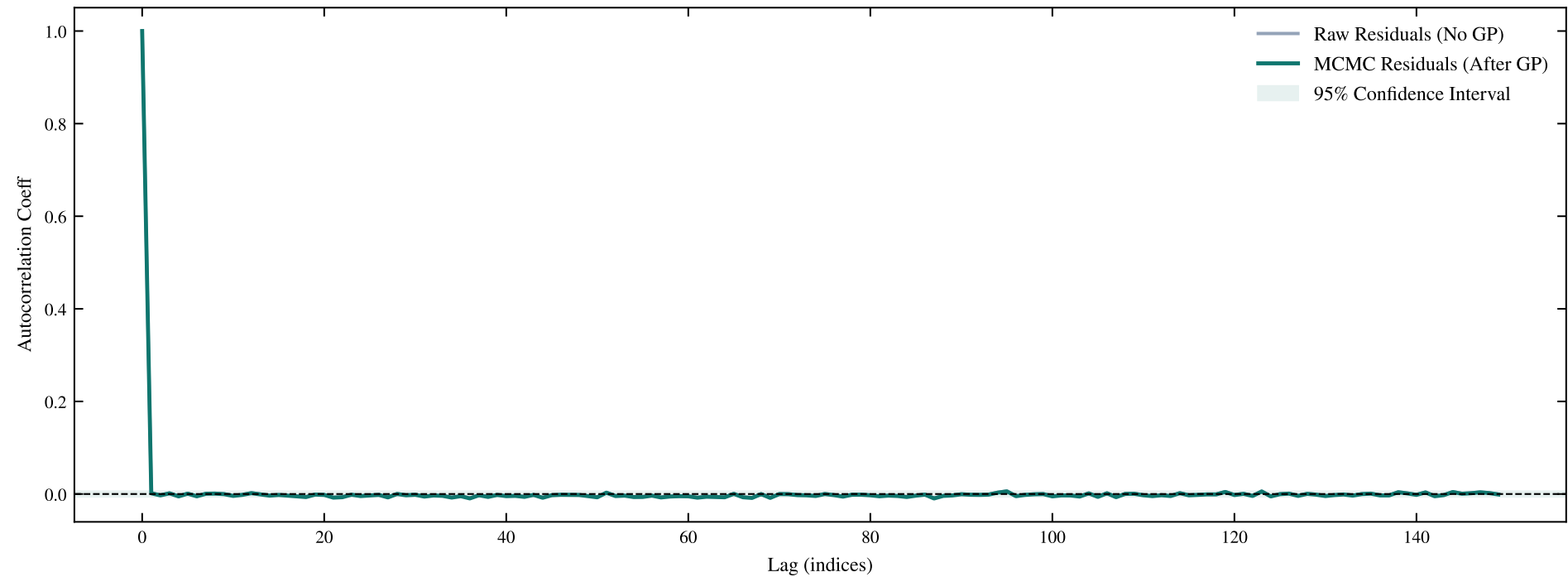
TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | Bayesian GP Detrending (Time Domain)



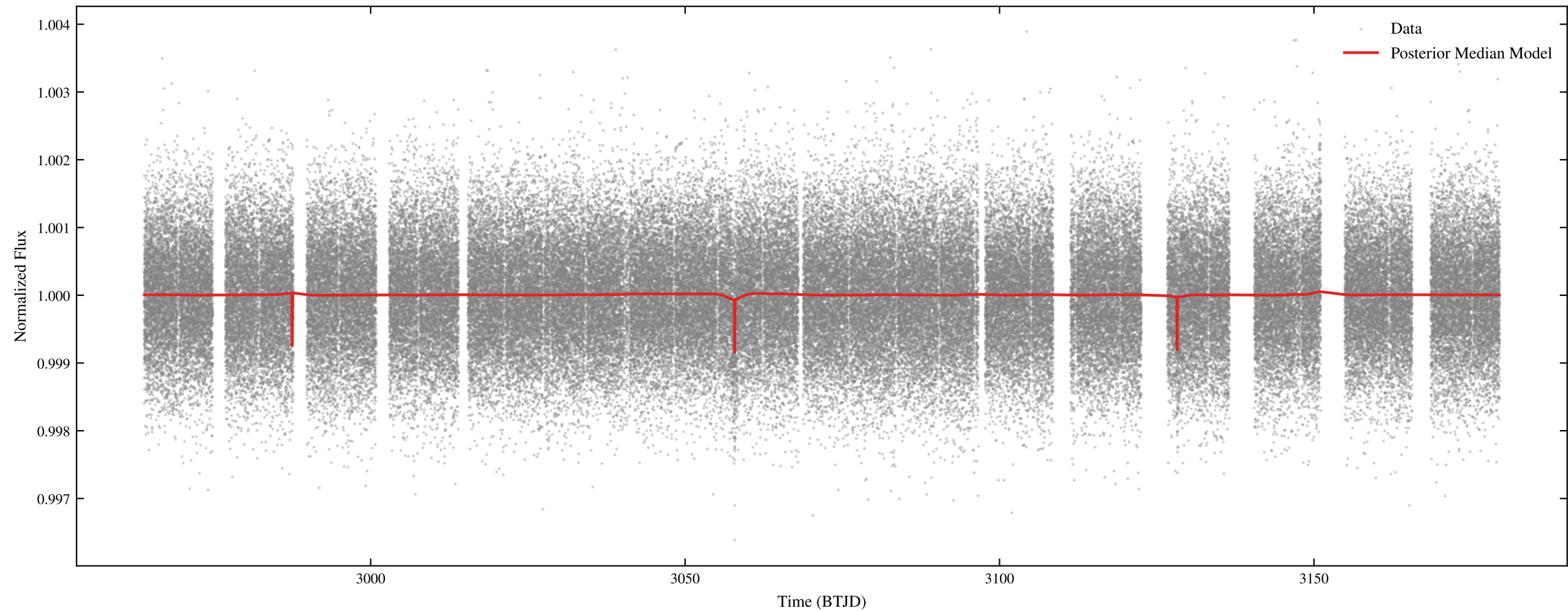
Detrended Light Curve & Best-Fit Transit Model

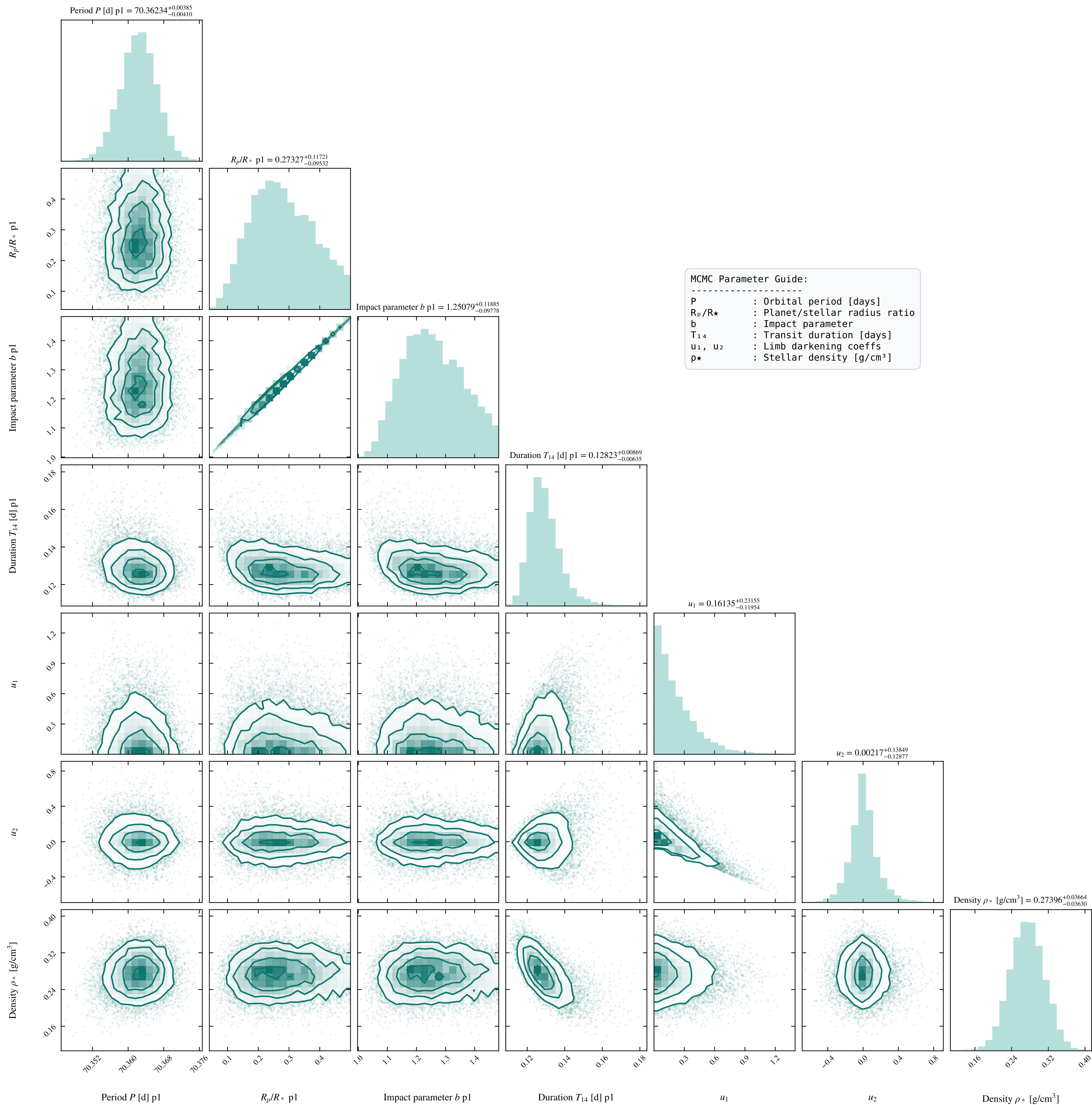


TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | GP Residuals Autocorrelation Function (ACF)

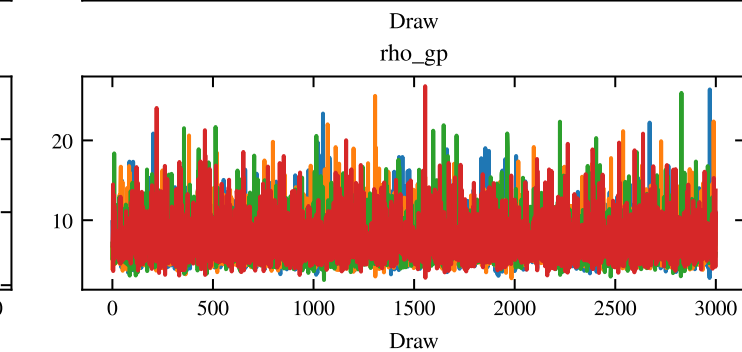
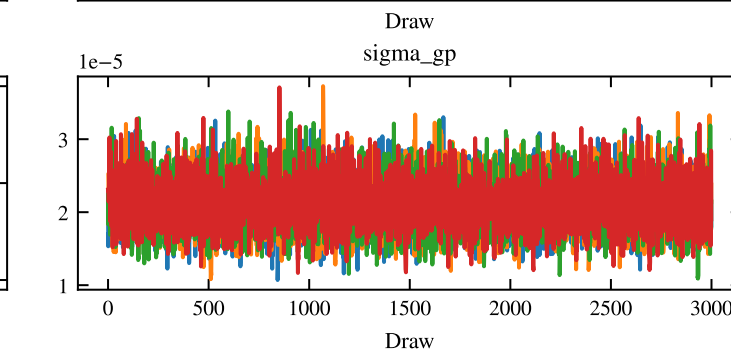
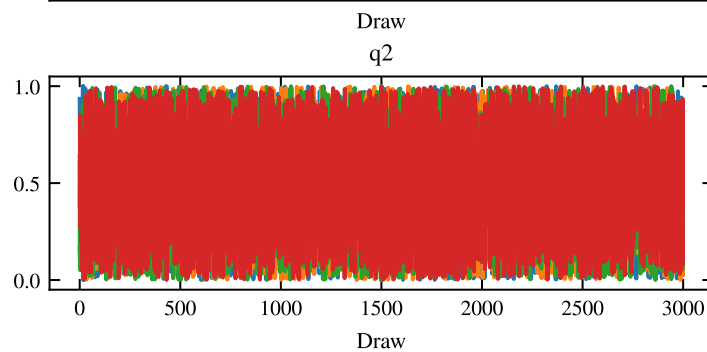
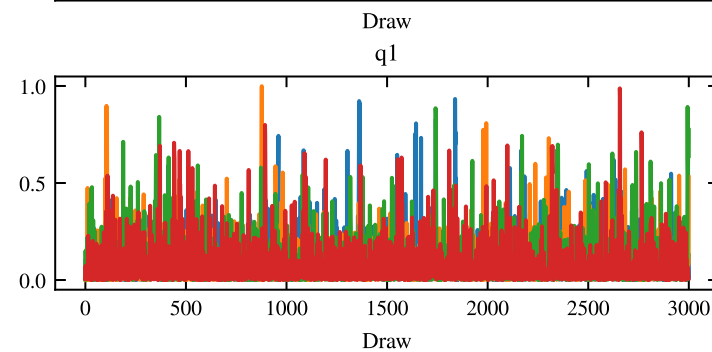
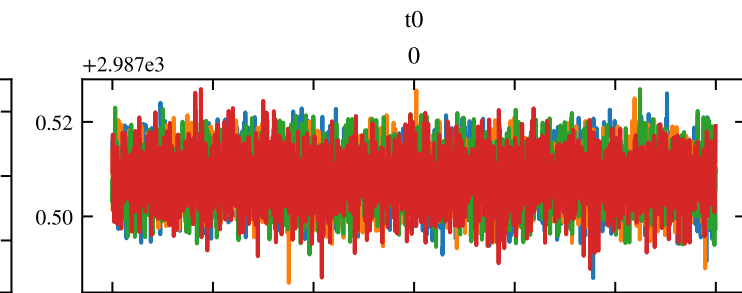
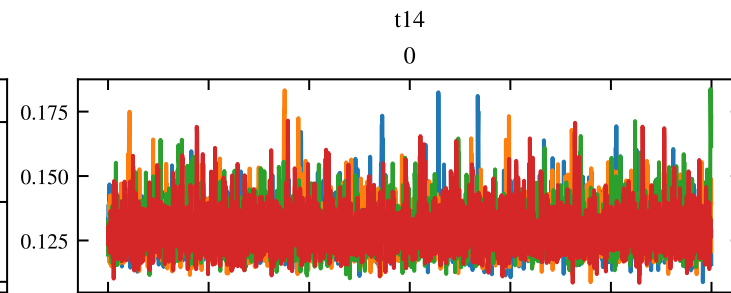
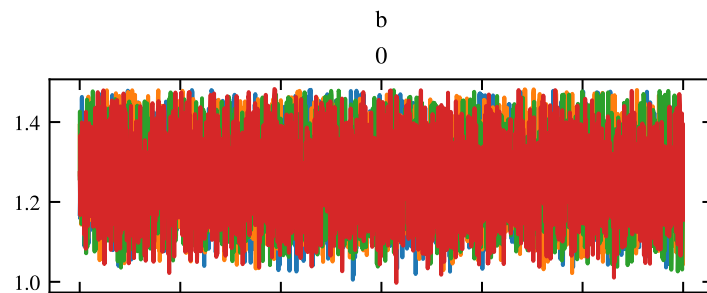
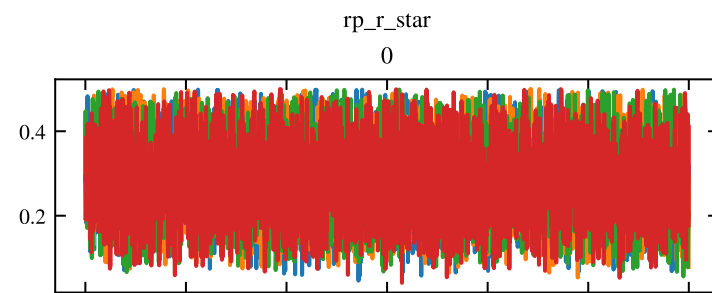


TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | MCMC Posterior Predictive Check

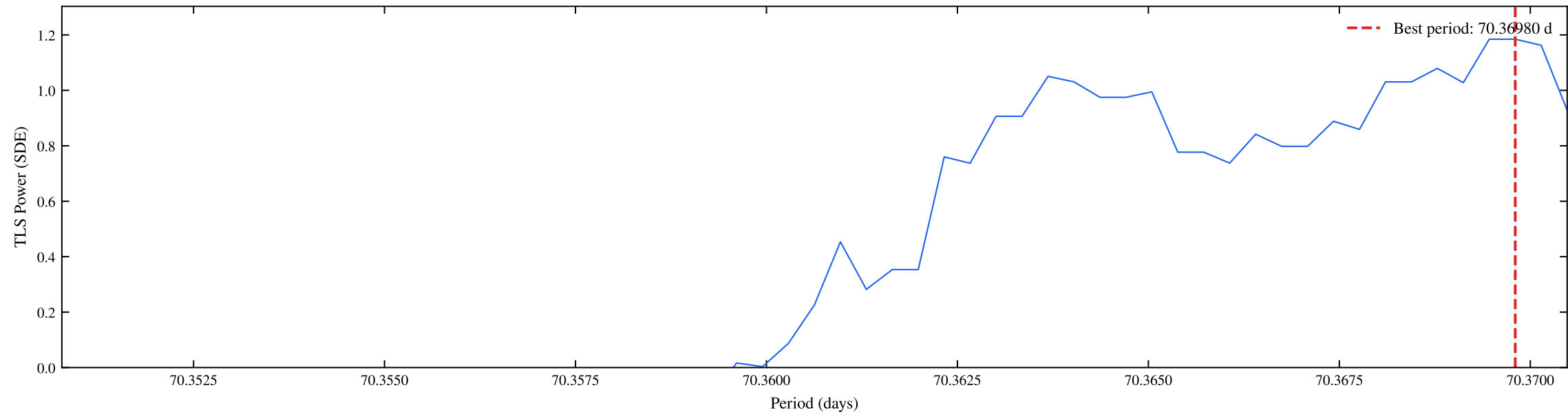




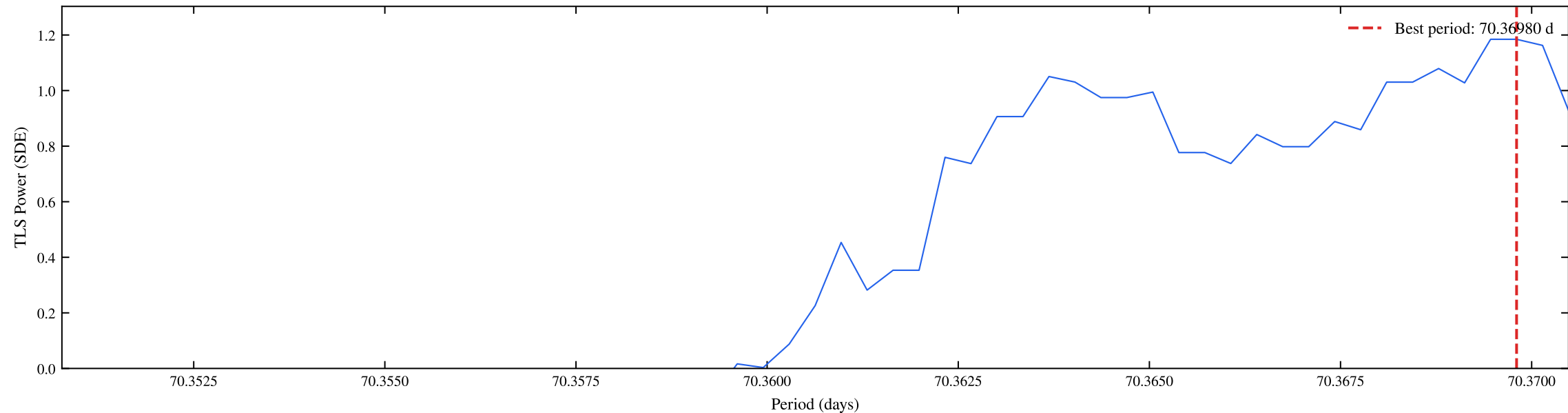
TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | MCMC Trace Plots



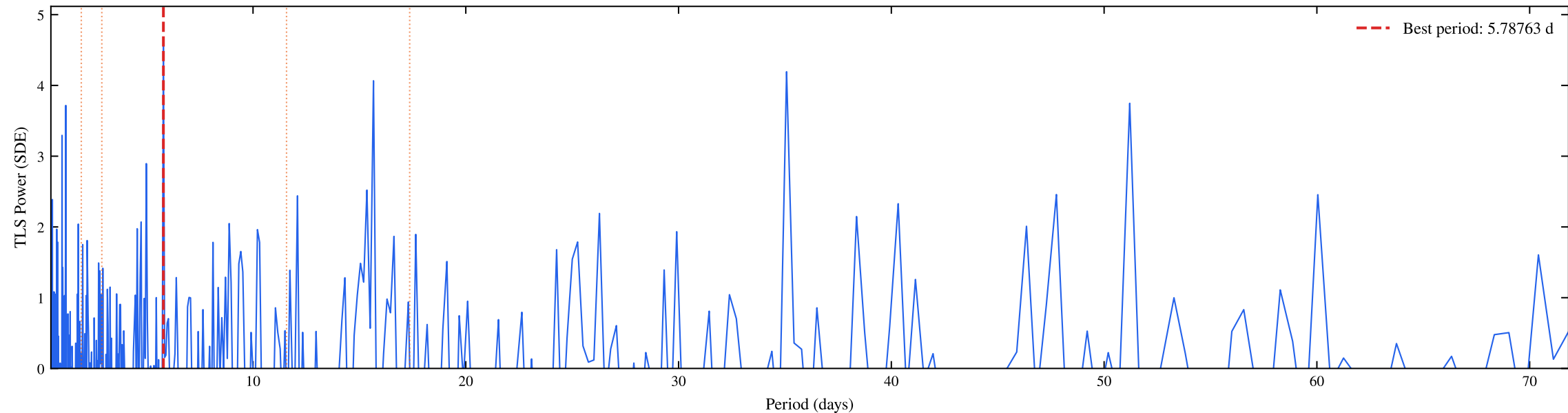
TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | TLS Coarse Periodogram (SDE = 1.18)



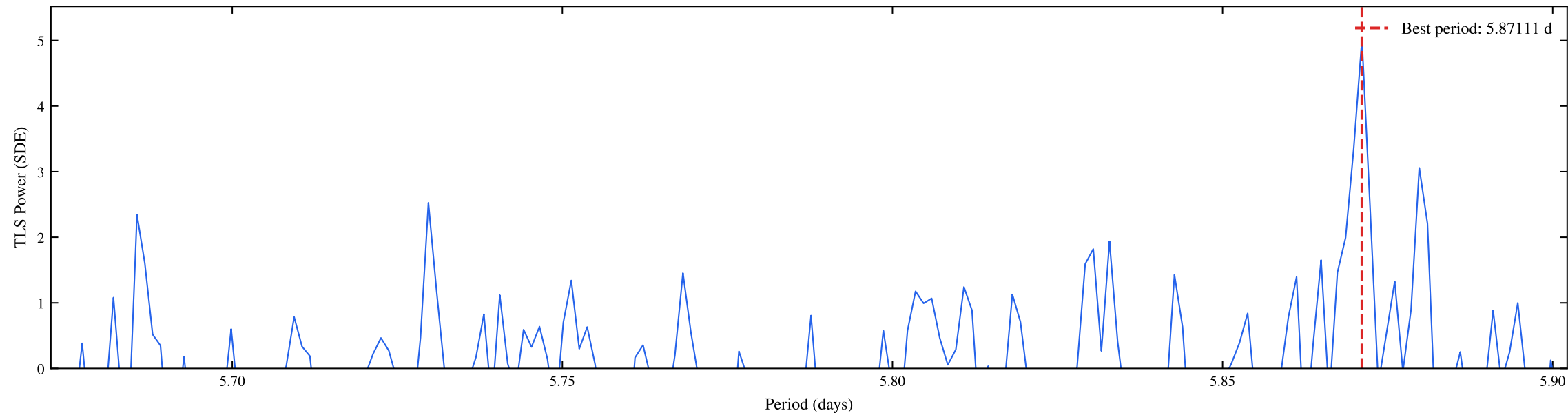
TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | TLS Fine Periodogram (SDE = 1.18)



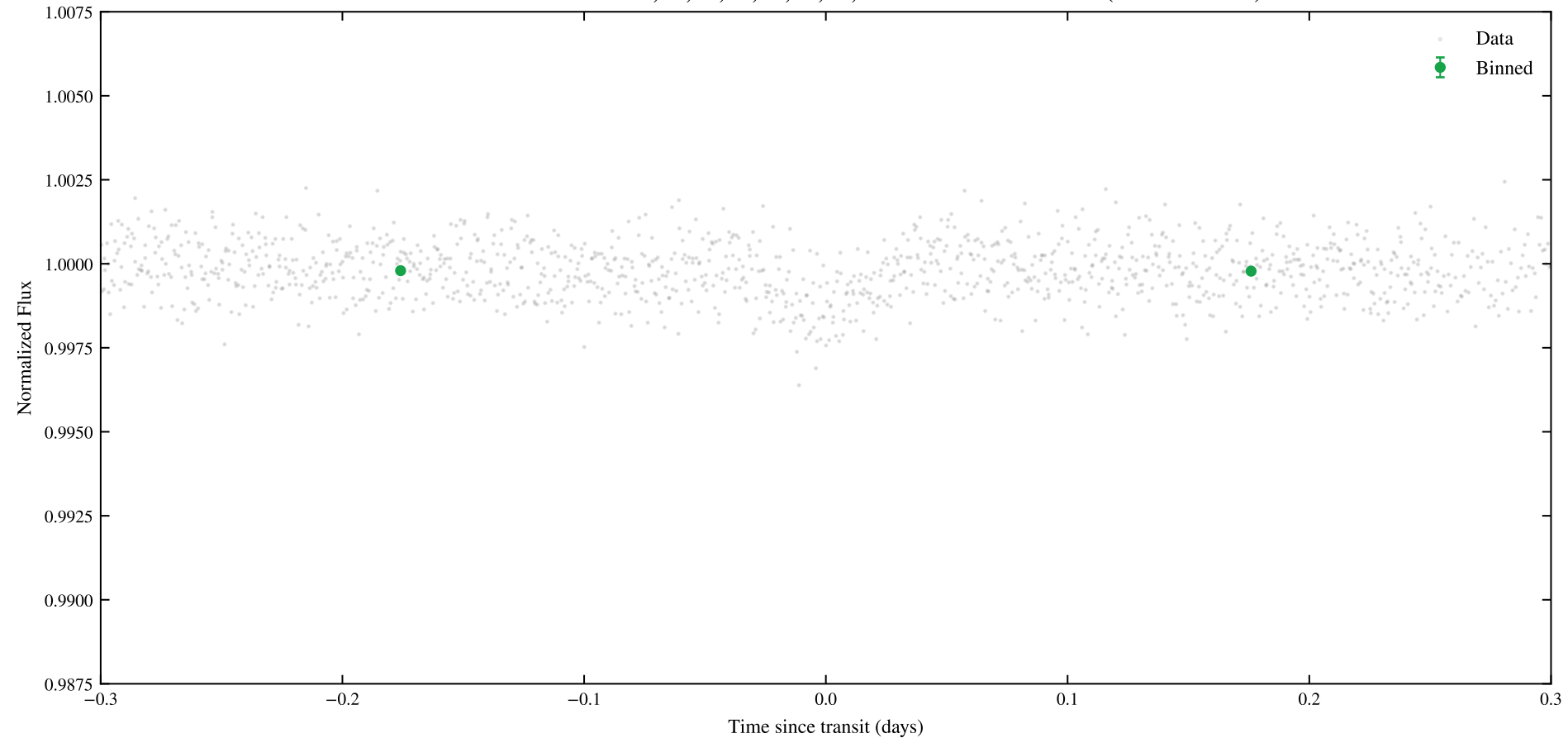
TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | TLS Coarse Periodogram (SDE = 4.65)



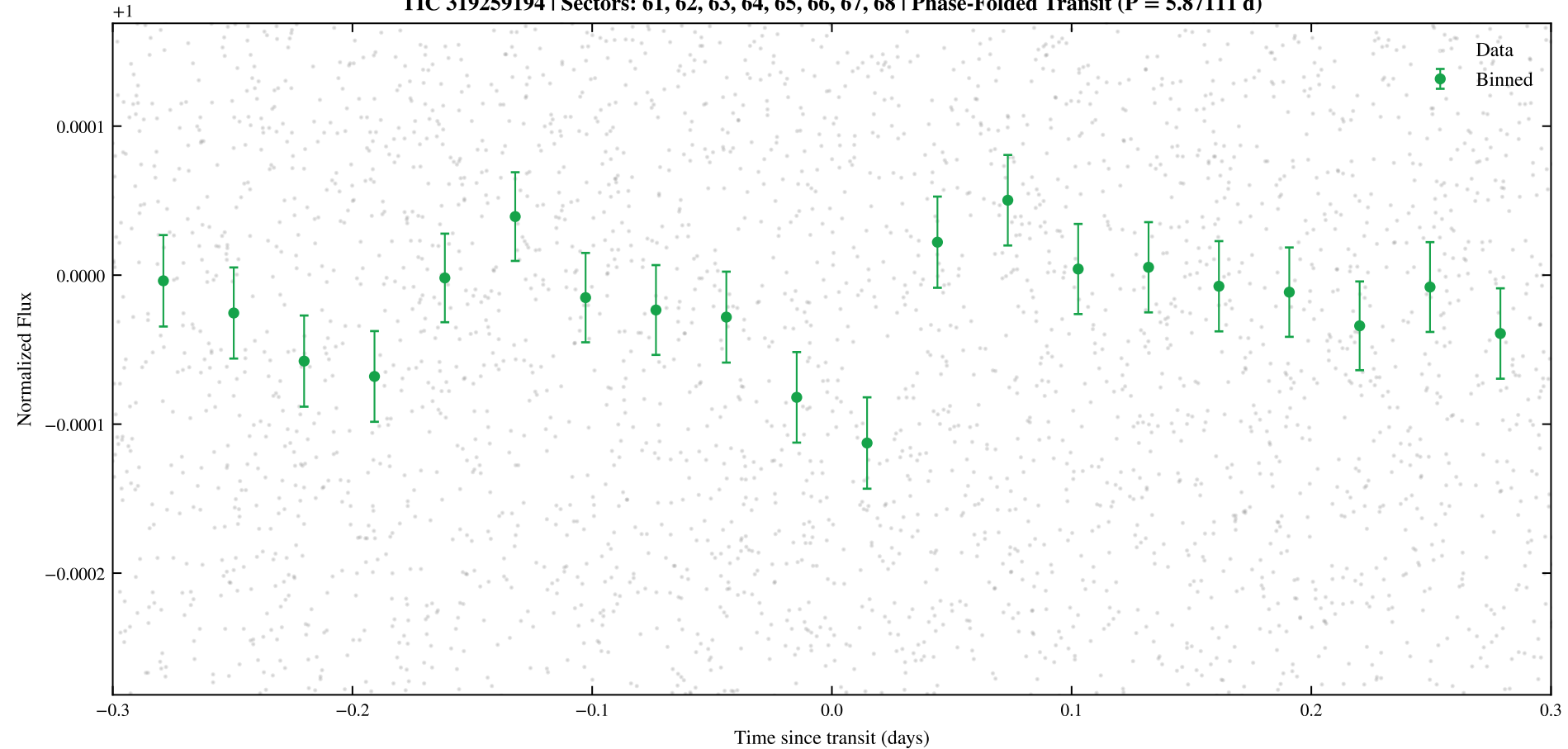
TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | TLS Fine Periodogram (SDE = 5.02)



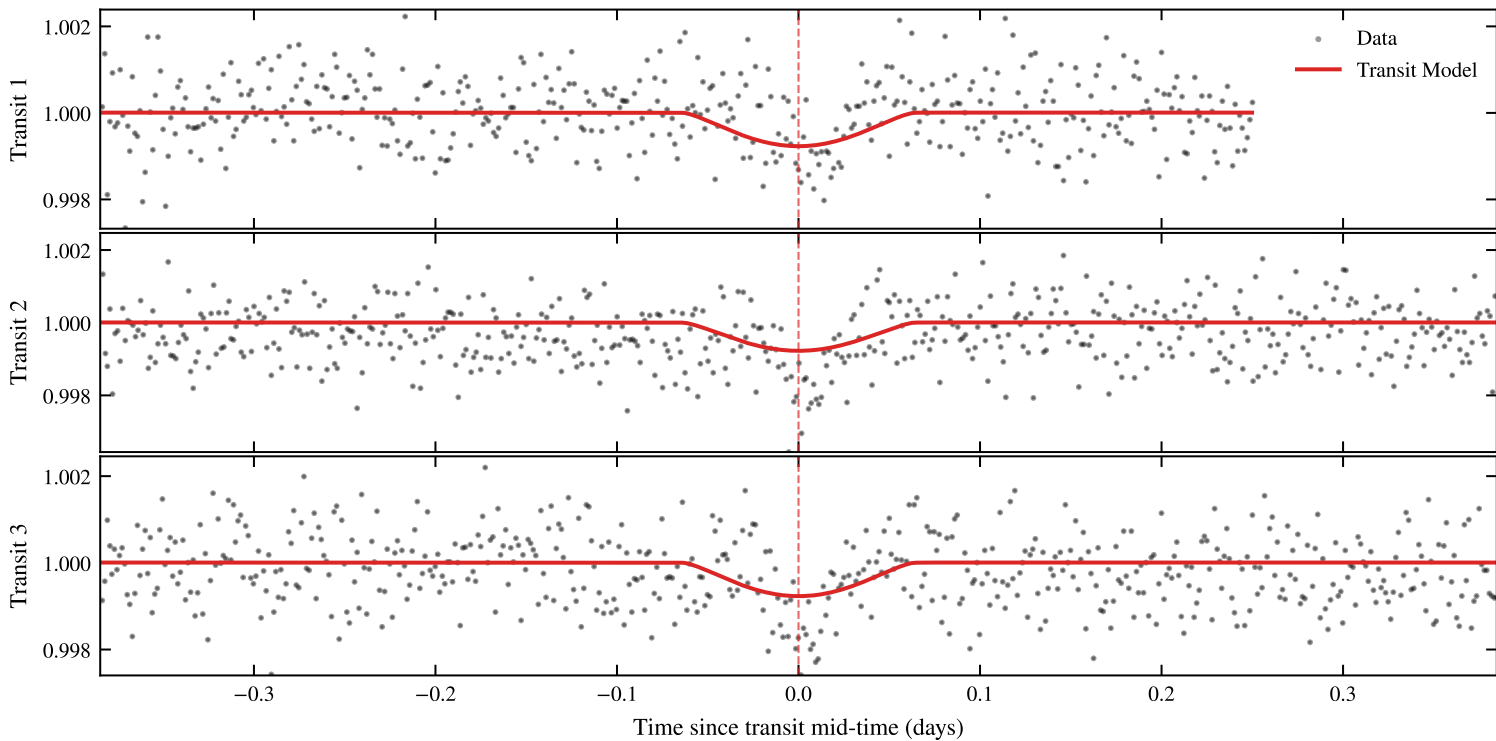
TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | Phase-Folded Transit (P = 70.36980 d)



TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | Phase-Folded Transit (P = 5.87111 d)



TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | Stacked Individual Transits



TIC 319259194 | Sectors: 61, 62, 63, 64, 65, 66, 67, 68 | Phased MCMC Fit - Planet 1

